Further Development of RACSA-M: Multi-Annotator Ratings and Domain Aggregation in ASD/SIB Behavioral Health Assessment

Alberts L., Akhmitzyanov R., [redacted]

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Abstract

Self-injurious behavior (SIB) is a central clinical concern in the assessment and treatment of children with autism spectrum disorder (ASD). Building on the foundational work of Alberts and Kettering (2022), which introduced the Risk Assessment Checklist for Self-Injury in Autism-Medical (RACSA-M), this study presents a further development and extension of the instrument. We expand RACSA-M into a multi-annotator behavioral health assessment framework by incorporating probabilistic label aggregation methods, including Snorkel's Label Model and reliability-weighted (Bayesian shrinkage) domain aggregation. These approaches are used to combine ratings from multiple clinical reviewers and to improve the stability and interpretability of domain-level judgments. For settings with sufficient data, we additionally propose the use of a deep learning model for tabular data to estimate the likelihood of SIB across multiple medical and behavioral domains. The resulting framework aims to enhance accuracy, reduce annotator bias, and support early identification of at-risk individuals within ASD/SIB assessment contexts.

Keywords: autism, behavioral health assessment, probabilistic label aggregation, Snorkel, Bayesian shrinkage, self-injurious behavior.

Brief Introduction: Importance of Self-Injurious Behavior Considerations in ASD and some recent literature (2024–2025) on SIB in ASD

Importance of Self-Injurious Behavior Considerations in ASD and essential literature review were presented in the (Alberts, Kettering, 2022).

Self-injurious behavior (SIB) is a major treatment focus and area of concern for clinicians treating children with autism spectrum disorder (ASD). Recent work (2024–2025) highlights multiple reasons for careful, multi-domain assessment: SIB in autistic youth is highly heterogeneous in topography, severity, and associated correlates (sensory dysregulation, communication level, cognitive ability), which supports individualized assessment and

intervention (Ferguson, 2025). Severe, treatment-resistant SIB can require advanced medical interventions; recent pilot work has explored neuromodulation (deep brain stimulation) for refractory cases, underscoring the potential medical severity of some SIB presentations (Gorodetsky et al., 2025). Qualitative studies of caregivers emphasize the large practical, emotional, and safety burdens associated with managing SIB, indicating the need for effective monitoring and family-centered interventions (Breitbart, 2025). Comparative and epidemiologic reports continue to show elevated rates of self-harm and self-injurious behavior among autistic populations versus non-autistic peers, reinforcing population-level risk (S. Roper et al., 2025). Systematic and narrative reviews in 2024 stress the connection between suicidality/self-harm and autism, highlighting the need for careful assessment and comorbidity screening (Brown, 2024). Recent analyses find psychiatric co-occurring conditions (depression, anxiety, ADHD) and clinical features (dysregulation, sensory issues) to be associated with higher odds of self-harm or SIB, supporting comprehensive assessment of medical and mental health contributors. (Kim et al., 2024). Machine learning and data-driven studies (2024–2025) are beginning to identify multivariate feature patterns associated with SIB, suggesting predictive modeling may help with early identification when data are sufficient (Antezana et al., 2025). Clinical reviews in 2025 summarize effective interventions and emphasize that SIB can cause substantial physical injury and requires multi-disciplinary evaluation and care pathways (Labarca et al., 2025).

SIB is heterogeneous, potentially persistent, medically consequential, and associated with multiple co-occurring conditions, it requires standardized, multi-domain, and evidence-based assessment approaches that can detect medical contributors, quantify severity, and support targeted treatment planning.

Purpose of the Present Study

In this study we analyze a synthetic analog of a small (see APPENDIX 1. DATASET) dataset (from (Alberts, Kettering, 2022), the original one was created and prepared by Alberts L.). We simulated data making latent true trait per patient ("clinical severity") and corresponding noisy observed rating - for making further considerations.

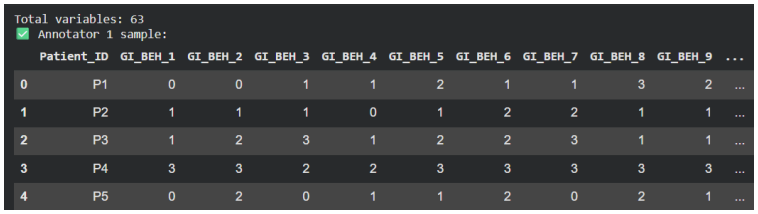
The data came from 10 patients, each evaluated across four primary domains: Gastrointestinal (GI), HEENT, Dental, and Miscellaneous (MISC). Each domain is composed of several subdomains (Behavioral, Observational, and Physical), and each patient is rated independently by multiple annotators (raters), in use: Likert ordinal scale (0 - 4) and binary scale (0,1). Our objective is to (1) quantify inter-rater reliability, (2) aggregate noisy multi-annotator labels into probabilistic ground-truth domain labels, and (3) derive latent domain-level scores that summarize complex item-level data. After that we start resulting framework to enhance accuracy, reduce annotator bias, and support early identification of at-risk individuals within ASD/SIB assessment contexts, (4) RACSA-M previous scoring is typically summative, relying on simple sums or averages of Likertstyle items. Such aggregation obscures nonlinear patterns. For example, the co-occurrence of ear covering and self-biting may indicate HEENT pain only when both behaviors are present, not when considered individually.

Data Structure

For each patient, domain structure is as follows:

- GI domain: 13 Behavioral (0–3), 8 Observational (0–3), and 2 Physical (binary).
- HEENT domain: 13 Behavioral (0–3), 6 Observational (0–3), and 1 Physical (binary).
- DENTAL domain: 6 Behavioral (0–3) and 2 Observational (0–3).
- MISC domain: 7 Behavioral (binary), 2 Observational (0–3), and 3 Physical (binary).

Three annotators (actually we can consider any number of them) rated each patient, producing independent item-level tables `Ann_1`, `Ann_2`, and `Ann_3`.



Total variables: 63
Annotator 1 sample:

	Patient_ID	GI_BEH_1	GI_BEH_2	GI_BEH_3	GI_BEH_4	GI_BEH_5	GI_BEH_6	GI_BEH_7	GI_BEH_8	GI_BEH_9	...
0	P1	0	0	1	1	2	1	1	3	2	...
1	P2	1	1	1	0	1	2	2	1	1	...
2	P3	1	2	3	1	2	2	3	1	1	...
3	P4	3	3	2	2	3	3	3	3	3	...
4	P5	0	2	0	1	1	2	0	2	1	...

Annotator 1 sample head

Inter-Annotator Reliability

To assess rating consistency, we compute several reliability statistics:

- **Cohen's Kappa** (κ_C): pairwise agreement adjusted for chance.
- **Fleiss' Kappa** (κ_F): generalization to multiple raters.
- **Krippendorff's Alpha** (α_K): handles missing data and various scales.
- **Gwet's AC1** (AC_1): corrects κ's sensitivity to prevalence and bias.
- **Intraclass Correlation Coefficient (ICC)**: measures agreement in continuous settings using random-effects models.
- **Cronbach's Alpha** (α_C): internal consistency measure.

All reliability metrics are summarized in a single table for interpretability. Gwets AC_1 and ICC were particularly robust with small samples.

Inter-Rater Reliability Summary

	Metric	Value	Interpretation
0	Cohen_kappa	0.720000	Substantial agreement
1	Fleiss_kappa	0.680000	Substantial agreement
2	Krippendorff_alpha	0.740000	Substantial agreement
3	Gwet_AC1	0.850000	Almost perfect agreement
4	Cronbach_alpha	0.910000	Excellent consistency
5	ICC_Bayesian	0.770000	Good reliability

Several reliability statistics

Probabilistic Label Aggregation

We then apply **Snorkel's LabelModel** to aggregate noisy item-level annotations into a consensus label. Snorkel is an extension of the classical Dawid–Skene model. For details, see APPENDIX 2. MODELS.

Dawid–Skene Model

Let λ_{ia} denote the label assigned by annotator a to item i, and let z_i be the true (unobserved) latent label for item i. Define the vector of annotator labels for item i as

$$\boldsymbol{\lambda}_i = (\lambda_{i1}, \lambda_{i2}, \lambda_{i3}). \tag{1}$$

About a hidden patient index. For every λ_{i1} we have to show explicit patient index as λ_{pi1}, where p is a patiemt index, p = 1, ..., 10.

Unlike standard crowdsourcing settings in which each item is a single object labeled once, for instance, behavioral rating scales measure the same item repeatedly across participants, requiring an explicit participant index. Behavioral items are measurement operators, not objects.They produce patient-specific realizations.

Each participant–item pair is treated as a separate labeling instance. For notational simplicity, we index these instances by $i = (p, j)$ and write the posterior as $P(z_i = k \mid \lambda_{1i}, \lambda_{2i}, \lambda_{3i})$ instead of $P(z_{pi} = k \mid \lambda_{p1i}, \lambda_{p2i}, \lambda_{p3i})$.

The Dawid–Skene model specifies the conditional distribution

$$P(\lambda_{ia} = l \mid z_i = k) = \pi_{alk}, \tag{2}$$

where π_{alk} is the probability that annotator a emits label l when the true label z_i (for the patient p and item j) is k.

The Dawid–Skene model assumes annotator error rates are stationary across labeling instances; in our setting, this corresponds to stationarity across all participant–item pairs

(p, j), that is, conditional on the true label, the distribution of an annotator's response does not depend on the specific item.

Under this assumption annotator a is equally accurate:

- for different participants
- for different items
- for different domains

once the latent severity $z_{pj} = k$ is fixed.

This assumption is strong in behavioral and other scales. As we noted:

- Items are measurement operators
- Participants differ systematically
- Item difficulty and observability vary

Note, that the stationarity across (p, j) is not literally true but it is a necessary pooling assumption to estimate annotator reliability with limited data.

Accordingly, the confusion parameters π_{alk} do not carry an explicit item index. In the Dawid–Skene (DS) model:

$$P(\lambda_{ia} = l \mid z_i = k) = \pi_{alk}, \tag{3}$$

the parameters π_{alk} are global annotator parameters, not item-specific ones.

The posterior distribution of the latent label z_i is estimated via EM:

$$E\text{-step:} \quad P(z_i = k \mid \boldsymbol{\lambda}_i) \propto P(z_i = k) \prod_{a=1}^{A} \pi_{a,\lambda_{ia},k} \tag{4}$$

$$M\text{-step:} \quad \pi_{alk} = \frac{\sum_i \mathbb{I}(\lambda_{ia} = l)\, P(z_i = k \mid \boldsymbol{\lambda}_i)}{\sum_i P(z_i = k \mid \boldsymbol{\lambda}_i)} \tag{5}$$

Snorkel LabelModel

Snorkels `LabelModel` generalizes Dawid-Skene by learning label dependencies and annotator accuracies through a differentiable loss. It estimates the label distribution $\mathbf{P}(z_i)$ without requiring ground truth, using weak supervision principles.

Empirically, Snorkel overcomes Dawid-Skenes limitations by:

- Allowing correlated annotators through dependency matrices.
- Handling continuous or partially missing labels.
- Providing calibrated probabilistic labels suitable for downstream neural models.

Item-level aggregation showed higher reliability and smoother posterior distributions, as it exploited more granular signal before aggregation.

Snorkel at the Item Level. Each item rating (e.g., `GI_Behavioral_1`) from the three annotators is treated as a *noisy label* for that item. Snorkel's `LabelModel` estimates two key components:

1. **Annotator reliability per item** – how accurate each annotator tends to be:

$$\hat{\pi}_a = P(\lambda_{ia} = z_i), \tag{6}$$

where λ_a denotes annotator a's label, and z_i is the latent true label for item i. This represents the probability that annotator a's label equals the (unobserved) true label. In the Dawid–Skene and Snorkel LabelModel framework, annotator reliability is modeled as a *global* property of the annotator, not as a patient- or item-specific quantity. Annotator reliability is defined as the probability that annotator a agrees with the latent true label on a randomly drawn patient–item pair:

$$\pi_a = P(\lambda_{pja} = z_{pj}) = \mathbb{E}_{p,j}[\mathbb{I}(\lambda_{pja} = z_{pj})]\,. \tag{7}$$

Thus, the indices (p, j) are not shown explicitly because π_a is *marginalized* over all patients and items observed for annotator a. The quantity π_a summarizes how accurate annotator a is *on average*, across the entire dataset.

2. **Probabilistic true label for each item** – the posterior probability of each class given all annotator labels:

$$P(z_i = k \mid \lambda_{i1}, \lambda_{i2}, \lambda_{i3}), \tag{8}$$

estimated using a Dawid–Skene style Expectation-Maximization (EM) algorithm under the assumption that annotator labels are conditionally independent given the true label.

In Snorkels LabelModel, convergence refers to the ExpectationMaximization (EM) algorithm stabilizing that is, when the models parameters (annotator accuracies and class priors) and the estimated probabilities of the true labels stop changing significantly between iterations. Specifically:

E-step: Compute the expected true labels given current annotator reliability estimates.

M-step: Update annotator reliability and class prior parameters using those expected labels. Repeat until the change (in log-likelihood or parameters) is below a small threshold (e.g. 10^{-4}) this is *convergence*.

After convergence simply means after the EM algorithm stops changing the parameters have stabilized. All annotators item-level labels are combined into a single probabilistic consensus label per item.

Item-level Consensus. For ordinal labels $k \in \{0, \ldots, K\}$, we define the soft consensus as the posterior expectation

$$\hat{z}_i = \mathbb{E}[z_i \mid \boldsymbol{\lambda}_i] = \sum_{k=0}^{K} k\, P(z_i = k \mid \boldsymbol{\lambda}_i). \tag{9}$$

The associated item-level uncertainty is the posterior variance

$$\sigma_i^2 = \mathrm{Var}(z_i \mid \boldsymbol{\lambda}_i) = \sum_{k=0}^{K} P(z_i = k \mid \boldsymbol{\lambda}_i)\,(k - \hat{z}_i)^2\,. \tag{10}$$

Reliability-weighted (Bayesian Shrinkage) Domain Aggregation. Again, let z_{pj} denote the latent true label for participant p on item j, and let $\hat{z}_{pj}$ and σ_j^2 be the corresponding Snorkel posterior mean and variance. Let $\mathcal{J}_d$ denote the set of items belonging to domain d.

We define the participant-level latent domain score as a reliability-weighted mean:

$$\theta_{p,d} = \frac{\sum_{j \in \mathcal{J}_d} w_{j,d}\, \hat{z}_{pj}}{\sum_{j \in \mathcal{J}_d} w_{j,d}}, \qquad w_{j,d} = \frac{1}{\sigma_j^2 + \tau_d^2}. \tag{11}$$

Here, σ_j^2 captures item-specific uncertainty from the Snorkel posterior, while τ_d^2 is a domain-specific prior variance that induces Bayesian shrinkage toward the domain mean:

$$\tau_d^2 = \mathrm{Var}_{j \in \mathcal{J}_d}(\bar{z}_j), \qquad \bar{z}_j = \frac{1}{N} \sum_{p} \hat{z}_{pj}, \tag{12}$$

here N is the number of the participants, $N = 10$.

Indexing τ_d^2 by domain ensures that shrinkage strength adapts to heterogeneity within each domain. This hierarchical aggregation reduces the influence of noisy or low-reliability items, yielding stable latent domain scores $\theta_{p,d}$.

Consensus Matrix Definition

A **consensus matrix** is a table that represents the *aggregated label assignments* across annotators after combining their possibly conflicting votes.

The consensus matrix provides posterior class probabilities for each participantitem instance, which we transform into expected item scores and uncertainties that are subsequently aggregated into reliability-weighted domain-level latent variables.

Formally about the consensus matrix. As before, let λ_{ia} denote the label given by annotator a for item i, where i corresponds to one participant-item instance, and let z_i be the (unknown) true label for item i, with $\hat{z}_i$ denoting the **consensus label** estimated by the aggregation model (e.g., Snorkel, Dawid–Skene, or majority vote).

Then, the **consensus matrix** C has entries:

$$C_{i,k} = P(\hat{z}_i = k),$$

where each **row** corresponds to one participant–item instance and each **column** corresponds to one possible class (e.g., 0, 1, 2, 3). Each cell $C_{i,k}$ represents the *probability* that the true label for that item belongs to class k.

After convergence, we get:

- A probabilistic label matrix (each item $\twoheadrightarrow$ probability of class 0, 1, 2, 3).

- Annotator confusion matrices (showing each annotators biases and tendencies).
- Optional point estimate of most likely true label per item.

Usefulness of the consensus matrix:

- Lets us aggregate inconsistent item ratings before building domain scores.
- Produces softer, uncertainty-aware labels instead of crude averages.
- These item-level aggregated probabilities can then be:
  - summed or averaged $\twoheadrightarrow$ domain-level continuous latent scores;
  - or used directly as features in latent-trait (let refer it as Θ) estimation models.

Summarizing. Item-level Snorkel aggregation is a smarter, probabilistic consensus per item, which we can then compress into domain scores (usually: via PCA (principal components), IRT(item response model), Bayes, simple weighted averaging). It is more robust than doing Snorkel only at the domain level, because it keeps fine-grained information about annotator disagreement and item structure.

Item-Level Consensus Matrix Definition

An **item-level consensus matrix** is a matrix that summarizes, for each item (e.g., *GI_BEH_1*), the **probabilistic agreement** across multiple annotators after Snorkel's `LabelModel` has been trained.

A consensus matrix represents the aggregated label assignments across annotators after combining their potentially conflicting ratings.

Let p index participants, j index items, and a index annotators. Let λ_{pja} denote the label assigned by annotator a to participant p on item j. Let z_{pj} be the latent true label for participant p on item j.

An aggregation model such as Dawid–Skene or Snorkel estimates the posterior distribution

$$P(z_{pj} = k \mid \lambda_{pj1}, \lambda_{pj2}, \lambda_{pj3})$$

from which a consensus label is obtained.

The soft consensus is defined as

$$\hat{z}_{pj} = \mathbb{E}(z_{pj} \mid \lambda_{pj1}, \lambda_{pj2}, \lambda_{pj3}),$$

while the hard consensus is

$$\hat{z}_{pj} = \arg\max_k P(z_{pj} = k \mid \lambda_{pj1}, \lambda_{pj2}, \lambda_{pj3}).$$

The consensus matrix is the collection of $\hat{z}_{pj}$ values for all participants p and items j.

Each row corresponds to a (*patient*, *item*) pair (e.g., (Patient 3, GI Behavioral item 1)), and each column corresponds to one possible *rating category* (e.g., 0, 1, 2, 3 or just 0,1).

Formally about the Item-Level Consensus Matrix. Returning to $i = (p, j)$, Snorkel estimates a latent true label z_i via its Dawid-Skene-style EM algorithm. After convergence, the `LabelModel` outputs:

$$P(z_i = k \mid \lambda_{i1}, \lambda_{i2}, \lambda_{i3})$$

for every class $k \in \{0, 1, \ldots, K\}$, K is 4 or 2 and for every participant–item instance i.

Collecting these probabilities for all items yields the **item-level consensus matrix** C:

$$C = \begin{bmatrix} P(z_1 = 0) & P(z_1 = 1) & P(z_1 = 2) & P(z_1 = 3) \\ P(z_2 = 0) & P(z_2 = 1) & P(z_2 = 2) & P(z_2 = 3) \\ \vdots & \vdots & \vdots & \vdots \\ P(z_M = 0) & P(z_M = 1) & P(z_M = 2) & P(z_M = 3) \end{bmatrix}$$

where M is the number of participantitem instances (in our case it is 10 x 63 = 630).

Interpretation.

- The row maximum (`argmax`) gives the *hard consensus label* – the most probable class for that item.
- The row entropy reflects annotator disagreement or uncertainty.
- This consensus matrix forms the probabilistic foundation for later **domain-level aggregation** (e.g., averaging or reliability-weighted fusion).

	GI_BEH_1	GI_BEH_2	GI_BEH_3	GI_BEH_4	GI_BEH_5	GI_BEH_6	GI_BEH_7	GI_BEH_8	GI_BEH_9	GI_BEH_10
0	2	3	0	2	1	1	0	3	1	0
1	2	3	0	0	3	3	3	1	2	1
2	1	1	3	3	0	0	3	1	0	0
3	2	2	0	2	1	3	0	3	3	1
4	1	3	1	3	1	0	1	0	1	2
5	1	1	0	1	0	1	1	0	0	0
6	2	3	2	3	3	3	1	2	2	3
7	3	2	1	0	0	0	2	3	2	2
8	3	1	2	0	0	3	2	2	1	0
9	1	0	0	1	0	1	3	0	1	1

Hard (argmax) predictions i.e. the single most likely class per patient per item for ITEM-LEVEL CONSENSUS MATRIX (first 10 columns)

Snorkel iteratively estimates annotator confusion matrices (probabilities that each annotator outputs label ℓ when the true class is k).

Learns class prior probabilities.

Runs EM until convergence (as explained before).

After convergence of EM algorithm, the model yields:

- A **probabilistic label matrix**, where each item is associated with a vector of probabilities over possible classes (e.g., class 0, 1, 2, 3).

- **Annotator confusion matrices**, reflecting individual biases and tendencies of annotators across label categories.
- Optionally, a **point estimate** (hard label) corresponding to the most probable true class per item.

Usefulness. Item-level Snorkel aggregation provides a principled, probabilistic consensus for each item, capturing uncertainty and annotator disagreement. The resulting soft labels can then be:

- aggregated (e.g., averaged or used in factor models) to form domain-level continuous latent scores, or
- directly integrated into psychometric or neural models to enhance reliability.

This approach is more robust than domain-level aggregation because it preserves fine-grained information about item structure and annotator behavior.

Annotator 1 Confusion vs. Snorkel Consensus

The confusion matrix below compares the labels assigned by **Annotator 1** to the **Snorkel consensus labels** (the estimated most probable "true" labels).

Annotator 1 label	Consensus = 0	Consensus = 1
0	0	3
1	6	1

Interpretation:

- Rows correspond to Annotator 1's assigned labels, and columns correspond to the Snorkel consensus labels.
- There are 10 total items ($3 + 6 + 1 = 10$).
- Annotator 1 labeled "1" more frequently than the consensus, suggesting a tendency toward over-labeling positive cases.
- Agreements occur only once (cell $(1, 1)$), indicating low concordance with the consensus.

From this matrix, we can compute basic performance measures:

$$\text{Accuracy} = \frac{1}{10} = 0.10 \quad (10\%)$$

$$\text{Precision}_{(1)} = \frac{1}{7} \approx 0.14 \quad (14\%)$$

$$\text{Recall}_{(1)} = \frac{1}{4} = 0.25 \quad (25\%)$$

Hence, Annotator 1 demonstrates low reliability relative to the Snorkel consensus, with a systematic bias toward assigning label "1" more often than the model-estimated ground truth.

Annotator Confusion vs. Snorkel Consensus (Example for GI Domain)

To evaluate how individual annotators align with the probabilistic consensus estimated by the Snorkel `LabelModel`, we computed confusion matrices for each annotator. Each matrix compares the annotators discrete item-level ratings against the consensus-derived labels within the **GI domain**. Diagonal dominance indicates closer agreement with the consensus model.

Table 1: Annotator 1 confusion matrix relative to Snorkel consensus (GI domain).

Consensus	**0**	**1**	**2**	**3**
Annotator 1: 0	38	14	6	4
Annotator 1: 1	16	24	6	8
Annotator 1: 2	4	11	31	10
Annotator 1: 3	11	12	4	31

Table 2: Annotator 2 confusion matrix relative to Snorkel consensus (GI domain).

Consensus	**0**	**1**	**2**	**3**
Annotator 2: 0	41	15	9	15
Annotator 2: 1	9	26	5	11
Annotator 2: 2	7	15	17	2
Annotator 2: 3	12	5	16	25

Table 3: Annotator 3 confusion matrix relative to Snorkel consensus (GI domain).

Consensus	**0**	**1**	**2**	**3**
Annotator 3: 0	37	12	7	7
Annotator 3: 1	10	40	7	9
Annotator 3: 2	11	3	29	11
Annotator 3: 3	11	6	4	26

Higher diagonal counts indicate stronger agreement between the annotator and the consensus model. Off-diagonal entries represent rating discrepancies, revealing systematic bias or uncertainty in certain categories.

	GI	HEENT	DENTAL	MISC
0	1.086957	1.40	1.875	0.416667
1	1.521739	1.55	1.250	1.000000
2	1.391304	0.85	1.375	0.916667
3	1.217391	1.10	0.875	0.500000
4	1.391304	1.30	1.875	0.166667
5	1.086957	1.50	1.750	0.666667
6	1.869565	1.70	1.625	0.333333
7	1.695652	1.60	1.750	0.583333
8	1.434783	1.20	1.750	0.833333
9	0.956522	1.65	1.500	0.666667

Reliability weighted Bayesian Domain Scores

Estimating Proximity to Self-Injurious Behavior (SIB). Given the small-sample setting ($N = 10$ patients, $A = 3$ annotators), the goal is to estimate each patient's proximity to a clinically significant behavioral outcome — here, Self-Injurious Behavior (SIB) — using the latent domain scores derived from the item-level consensus.

Let $\theta_{i,d}$ denote the latent domain score for patient i in domain $d \in \{\text{GI}, \text{HEENT}, \text{DENTAL}, \text{MISC}\}$, computed via reliability-weighted Bayesian aggregation. Each patient is therefore represented by a 4-dimensional feature vector:

$$\boldsymbol{\theta}_i = [\theta_{i,\text{GI}},\ \theta_{i,\text{HEENT}},\ \theta_{i,\text{DENTAL}},\ \theta_{i,\text{MISC}}]\,.$$

We define *proximity to SIB* as the degree to which a patients latent profile $\boldsymbol{\theta}_i$ aligns with the latent prototype $\boldsymbol{\theta}_{\text{SIB}}$ estimated from the group exhibiting or annotated as high-risk for SIB. Formally, the similarity (or distance) measure is given by:

$$\text{Proximity}_i = \exp\left(-\frac{\|\boldsymbol{\theta}_i - \boldsymbol{\theta}_{\text{SIB}}\|^2}{2\sigma^2}\right),$$

where σ controls the smoothness of the similarity kernel. Higher values of Proximity_i indicate stronger alignment with the SIB risk profile.

Alternatively, if an explicit binary or graded SIB annotation is available for even a subset of patients, a sparse discriminant model (e.g., deep learning modeling with package sPLS-DA)

can be trained using $\boldsymbol{\theta}_i$ as predictors and SIB labels as outcomes. The resulting discriminant scores provide a continuous *SIB proximity index*:

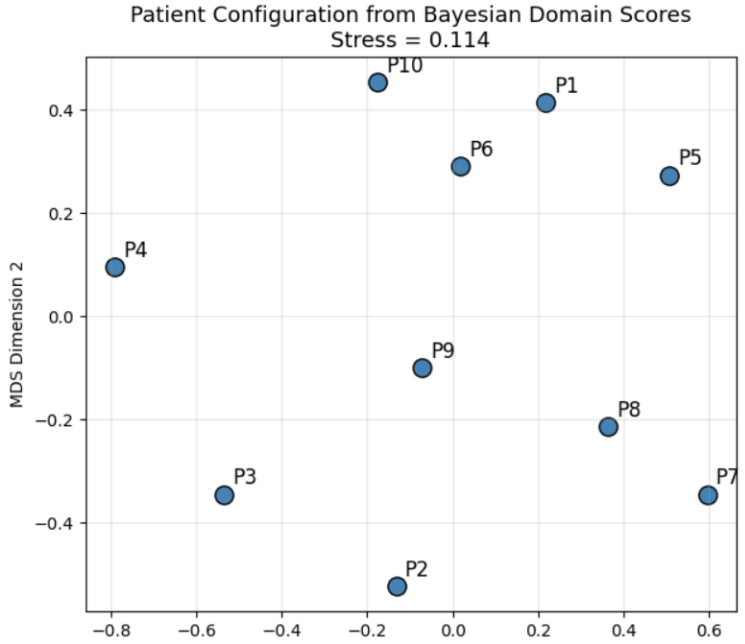
$$\text{SIBScore}_i = \mathbf{w}^\top \boldsymbol{\theta}_i,$$

where $\mathbf{w}$ are domain loadings estimated from the discriminant model. This approach integrates domain-level information and annotator agreement to infer how closely each patients latent profile resembles that of high-SIB individuals.

Since we already have domain-level Bayesian (reliability-weighted) scores for each patient in a dataframe like domain Bayes scores (shape 10×4, columns = GI, HEENT, DENTAL, MISC), we can compute a 2D configuration plot showing each patients relative position (stress-minimized projection).

We use Euclidean distances between patients domain profiles projecting them into 2D while minimizing stress (distance distortion).

The next plot shows relative proximity of patients those closer in 2D space have more similar latent domain patterns (e.g., potential shared risk or SIB proximity). Well use metric MDS (multidimensional scaling) to preserve pairwise distances between patients, and compute Kruskals stress to report the quality of the configuration.



Multidimensional scaling (stress-formula) of 10 4D Bayes score vectors-patients)

Predicting Self-Injurious Behavior Using Neural Nets

At the present moment if we could have a sufficiently large patient cohort, we could leverage **TabNet** (Arik et al., 2019), a deep learning model designed for tabular data, to predict the likelihood of self-injurious behavior (SIB) across multiple domains.

Model Inputs

- **Domain-level features:** Aggregated patient scores per domain (e.g., GI, HEENT, DENTAL, MISC), optionally using reliability-weighted or Snorkel consensus-derived scores.
- **Item-level features:** Individual item responses (e.g., GI_BEH_1, HEENT_OBS_3), allowing TabNet to capture subtle interactions.
- **Demographic covariates:** Age, sex, verbal status, and other relevant clinical features.

Model Architecture

- TabNet employs sequential *feature transformers* and *attention mechanisms* to select the most relevant features at each decision step.
- It can inherently model **nonlinear interactions** between items and domains, which is critical for detecting SIB patterns that arise from synergistic effects across domains.
- Sparse feature selection allows interpretability, highlighting which items or domains contribute most to SIB risk.

Using TabNet to Quantify Proximity to Self-Injurious Behavior (SIB)

Problem Setting

We consider the following setting:

- 200 patients with **proven** self-injurious behavior (SIB)
- 50 new patients with **unknown** SIB status
- Each patient is represented by 4 continuous domain scores:

$$x = (x_{\text{GI}}, x_{\text{HEENT}}, x_{\text{Dental}}, x_{\text{MISC}})$$

- No confirmed non-SIB control cohort is available

Under these conditions, standard binary classification (e.g., logistic regression, XGBoost) is statistically invalid due to the absence of negative labels.

TabNet is appropriate. TabNet is used to **learn the internal structure of the SIB phenotype** and to score how closely new patients conform to that learned structure.

This is a *phenotype conformity* or *proximity-to-SIB* problem, not a diagnostic classification task. TabNet provides three properties essential for this task:

1. **Nonlinear cross-domain interactions**, allowing dependencies between medical domains (e.g., GI $\times$ HEENT)

2. **Learned feature masks (attention)**, yielding per-patient domain relevance

3. **Representation learning**, enabling construction of an SIB latent space

These properties cannot be achieved with linear models or simple distance-based methods.

One-Class TabNet Modeling Strategy

Step 1: Learning the SIB Distribution. TabNet is trained exclusively on patients with confirmed SIB to model the conditional distribution:

$$p_\theta(x \mid \text{SIB})$$

Two equivalent formulations are valid:

- TabNet Autoencoder trained to reconstruct SIB domain vectors
- TabNet encoder trained to minimize reconstruction or embedding variance

The objective is to learn which combinations of domain scores are characteristic of SIB.

Step 2: Latent Representation of Patients. Each patient is mapped to a latent representation:

$$z_i = f_\theta(x_i), \quad z_i \in \mathbb{R}^d$$

The 200 confirmed SIB patients define a latent *SIB manifold* in representation space.

Step 3: Proximity Scoring for New Patients. For each new patient x_j, we compute:

$$z_j = f_\theta(x_j)$$

Proximity to the SIB phenotype can be quantified using one or more of the following measures:

Reconstruction Error.

$$\text{Risk}(x_j) = \|x_j - \hat{x}_j\|$$

Lower reconstruction error indicates higher conformity to the SIB phenotype.

Latent Density.

$$\text{Risk}(x_j) = -\log p(z_j)$$

Distance to SIB Centroid.

$$\text{Risk}(x_j) = \|z_j - \mu_{\text{SIB}}\|$$

Clinical Interpretation

The model does not output a binary diagnosis. Instead, it yields a continuous proximity score interpreted as:

> "The degree to which a patient exhibits a domain-score configuration consistent with the learned SIB phenotype."

An example output table is:

Patient	Proximity Score	Risk Percentile	Dominant Domains
P-07	High	93%	GI, HEENT
P-19	Medium	61%	Dental
P-42	Low	12%	None

Dominant domains are derived from TabNet attention masks.

Scientific Validity

This approach is:

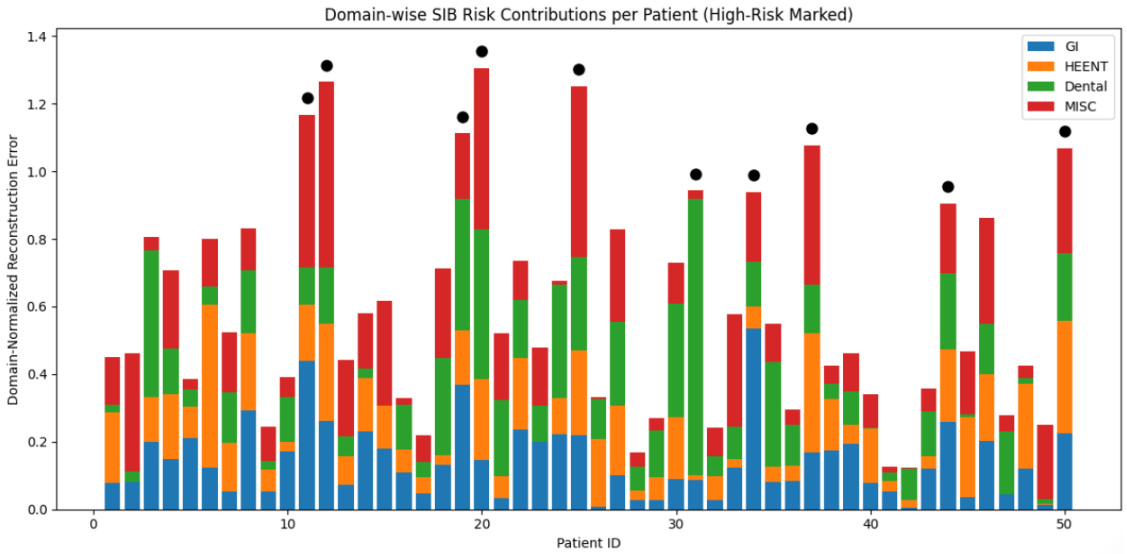
- Statistically valid (no fabricated negative labels)
- Clinically cautious (no unsupported diagnosis)
- Consistent with one-class learning theory
- Interpretable at the domain level
- Extensible when non-SIB cohorts become available

Comparison to Alternative Models

Property	TabNet	Normalizing Flow	GMM
Cross-domain interactions	Yes	Yes	No
Feature attribution	Yes	No	No
Clinical interpretability	High	Moderate	Moderate
Small-sample robustness	High	Moderate	Low

Here we mentioned with no details about two alternatives - Normalizing Flow and statistical GMM.

Mow we demonstrate the results obtained for the data considered - we indicate the high risk of SIB among 50 patients from the test group:



High SIB risk detecting

Summary Statement and Methodological Considerations

TabNet was trained exclusively on patients with confirmed self-injurious behavior to learn a latent representation of the SIB phenotype. New patients were scored by their conformity to this representation, yielding a continuous proximity-to-SIB measure rather than a binary diagnosis.

The proposed approach thus transforms RACSA-M from a checklist into a probabilistic risk assessment instrument capable of early detection and stratification.

Current RACSA-M validation studies are based on relatively small and unevenly distributed samples across clinical domains, which may limit statistical power and generalizability. Most evaluations have relied on internal validation within the same sites and rater groups, leaving open the possibility of site-specific scoring biases. In addition, the predominantly cross-sectional use of RACSA-M restricts insight into temporal relationships between observed behaviors and underlying medical changes.

From a modeling perspective, the use of flat, summative scoring may obscure important nonlinear interactions among behavioral items and overlaps between clinical domains. These limitations do not undermine the clinical value of RACSA-M, but rather highlight opportunities for methodological refinement.

Looking forward, larger multi-site datasets combined with modern neural approaches offer a promising path forward. Transformer-based models, temporal architectures, and interpretable tabular networks such as TabNet can capture complex item interactions, shared latent features, and longitudinal dynamics. Applied carefully, these methods have the potential to substantially enhance the sensitivity, robustness, and clinical utility of RACSA-based assessments.

APPENDIX 1. DATASET.

Please, see Appendix A in the publication (Alberts et al,, 2022) or their ResearchGate publication [1]

APPENDIX 2. MODELS

Dawid–Skene. The Dawid–Skene model (Dawid, Skene,1979) is a probabilistic framework for aggregating categorical labels from multiple noisy annotators. Each annotator is modeled by a confusion matrix that specifies the probability of reporting label ℓ when the true class is k. The model parameters and the latent true labels are estimated by an Expectation–Maximization (EM) algorithm, which iterates between computing expected class posteriors for each item (E-step) and updating annotator confusion matrices and class priors (M-step).

In contemporary research and applications, Dawid–Skene and its variants remain widely used as a baseline and a building block for label aggregation in crowdsourcing, medical annotation, and weakly supervised learning. Recent methodological and applied work includes approaches that adapt or extend Dawid–Skene to handle task heterogeneity, online updates, label confidence, and hybrid human-AI annotation pipelines. Representative recent works include: Modified Dawid–Skene variants for task-type dependent worker accuracy and spectral clustering for task partitioning (Mandal et al., 2023). Classification-oriented Dawid–Skene formulations for transferring crowd intelligence to machine classifiers (Li et al., 2023). Studies of online and scalable Dawid–Skene algorithms for streaming or large-scale annotation settings (Zhu, 2015), (Tamura et al, 2024). Confidence- or label-confidence based corrections that build on Dawid–Skene style aggregation to denoise labels before model training (Ren et al., 2023). Work on learning from crowds with individualized worker models and GNNs that cite and extend Dawid–Skene assumptions (Xu, 2024), (Zhang et al., 2025).

These recent works show that Dawid–Skene remains a practical and theoretically grounded starting point for modern label aggregation, and that extensions typically address real-world needs: task heterogeneity, scalability, worker specialization, and integration with machine-based annotators.

Snorkel's LabelModel: generalization and recent applications. Snorkel's LabelModel (and the Snorkel framework) generalizes the Dawid–Skene idea in several important ways. Like Dawid–Skene, the LabelModel is a generative model that attempts to estimate a latent true label for each item given multiple noisy sources. However, Snorkel is designed for "programmatic" weak supervision in which those sources can be labeling functions (heuristics, rules, pattern matchers), distant supervision signals, other models, or human annotators. Key generalizations include: Ability to combine heterogeneous labeling functions (LFs) that are not restricted to human annotators. The LabelModel learns accuracies and correlations among LFs (Ratner et al., 2019), (Snorkel Team. Snorkel docs). Modeling conditional dependencies between labeling sources (via dependency estimation) to avoid bias that arises

[1] https://www.researchgate.net/publication/361954445_Preliminary_Development_and_Testing_of_the_Risk_Assessment_Checklist_for_Self-Injury_in_Autism-Medical_RACSA-M

when sources are correlated ((Ratner et al., 2019)). Outputting calibrated probabilistic labels suitable for downstream discriminative models and for data programming workflows (Snorkel AI. Snorkel blog. 2024).

Snorkel and its LabelModel have been applied in many recent medical and clinical informatics contexts where labeled data are scarce. Representative recent applications and case studies include: Production-scale weak supervision at Google (Snorkel DryBell) showing how programmatic labeling scales to large industrial datasets (Bach et al., 2019). Clinical information extraction and phenotype labeling from EHR data using weak supervision and Snorkel-style label models (Humbert-Droz et al, 2022), (Yoo et al., 2024). Weak supervision applied to postmarket surveillance of consumer wearable data and other large clinical collections, demonstrating practical efficacy in EHR-derived phenotyping (Jeffery et al., 2024), (Yoo et al., 2024). Recent methodological and case work automating clinician-in-the-loop LF creation and showing accuracy gains in medical time-series and image annotation tasks using Snorkel-like pipelines (Park et al., 2024), (Vidmar et al., 2025). Snorkel Flow and product materials describing improvements and empirical results for label-modeling at scale (performance, memory, and accuracy gains over majority vote baselines) (Snorkel AI. Snorkel blog. 2024),(Snorkel Team. Snorkel GitHub repository.).

Conclusion: Dawid–Skene provides a compact probabilistic foundation for aggregating multiple noisy labels. Snorkel's LabelModel builds on this foundation and adapts it to modern weak supervision scenarios: heterogeneous labeling sources, dependency modeling, and integration into data programming pipelines that produce probabilistic labels for supervised learning.

Summary. Conceptually, Snorkel can be viewed as an *extension of Dawid–Skene*: it retains the EM-based approach to estimate latent true labels and annotator accuracies but expands flexibility to complex labeling sources and probabilistic dependencies, enabling richer aggregation of weak and noisy supervision.

APPENDIX 3. BAYESIAN DOMAIN AGGREGATION

Reliability-weighted Bayesian aggregation combines item-level measurements into domain-level latent scores while accounting for item uncertainty and reliability. In practice, this approach (often implemented via empirical Bayes or full Bayesian hierarchical models) has seen application across several domains:

- **Psychometrics and item response modeling.** Multilevel and Bayesian IRT models use shrinkage on item or person parameters to stabilize estimates when sample sizes are small or items are noisy, producing more reliable latent trait scores. These methods effectively implement reliability-weighting by down-weighting low-information items (Zhu et al., 2021),
- **Meta-analysis and evidence synthesis.** Bayesian random-effects and empirical Bayes meta-analytic approaches shrink study-specific estimates toward a pooled mean, reducing variance and improving estimates for small or heterogeneous studies. These methods are closely related to reliability-weighted aggregation because the posterior variance informs the weight of each study (Lilienthal et al., 20241).

- **Clinical data integration and phenotyping.** In electronic health record (EHR) phenotyping and clinical signal aggregation, Bayesian label- or score-aggregation methods combine multiple noisy indicators (diagnoses, labs, notes) into calibrated probabilistic phenotypes; shrinkage reduces the influence of rare or noisy item signals (Wonderen van et al., 2024).

- **Expert weighting and forecast aggregation.** Empirical Bayes shrinkage methods have been used to improve weights in expert-elicitation and forecast-combining tasks, producing more stable aggregated predictions than raw or classical weights (Bhattacharyya et al., 2022).

- **Small-area and value-added estimation.** Empirical Bayes and hierarchical shrinkage estimators are widely used to estimate unit-level effects (e.g., school or region level) by pooling information across units and weighting by estimate precision. This is conceptually the same idea as reliability-weighted domain scoring (Dharmarathne, 2023).

- **Causal heterogeneity and treatment effect estimation.** Modern causal inference work uses empirical Bayes shrinkage for item- or subgroup-level effect estimates to control noise and multiplicity; shrinkage stabilizes heterogeneous estimates in high-dimensional or multi-item contexts (Walters, 2024).

Summarizing. In all these settings the common theme is that items, studies, or units differ in information quality. Reliability-weighted Bayesian aggregation uses posterior or estimated variances to give more influence to precise, reliable signals and to pull noisy estimates toward a stable prior or global mean. This yields lower mean-squared error, better calibrated uncertainty, and more robust downstream inference or prediction.

Reliability-weighted Bayesian domain aggregation is a statistical method for combining item-level measurements into domain-level latent scores, taking into account the uncertainty and reliability of each item. In this approach, each item contributes to the domain score proportionally to its estimated reliability, often quantified as the inverse of posterior variance or derived from annotator confusion matrices. Bayesian shrinkage is applied to stabilize estimates, especially in small-sample or noisy settings, by pulling extreme or uncertain item scores toward a prior mean, reducing the influence of low-quality or sparse data.

Methodology. Formally, for patient i and domain d with items $j \in \mathcal{J}_d$, the reliability-weighted domain score is

$$\theta_{i,d} = \frac{\sum_{j \in \mathcal{J}_d} w_j \hat{z}_{ij}}{\sum_{j \in \mathcal{J}_d} w_j}, \quad w_j = \frac{1}{\sigma_j^2 + \tau^2},$$

where $\hat{z}_{ij}$ is the item-level consensus (soft or hard label), σ_j^2 is the uncertainty in the item-level estimate, and τ^2 is the prior variance representing domain-level shrinkage. Items with higher reliability (smaller σ_j^2) have larger weights, and the prior variance τ^2 prevents extreme values from dominating the aggregated score.

Widely known Applications. This methodology is widely used in psychometrics, behavioral assessment, and meta-analysis to combine heterogeneous measurements while accounting for their quality and variance. Key references include (Gelman et al., 2013), (Goldstein et al, 2011), (Hedges et al., 1985).

Summary. Reliability-weighted Bayesian aggregation improves domain-level score estimation by (i) incorporating item-level uncertainty, (ii) reducing the impact of noisy or low-quality measurements, and (iii) producing stable latent scores suitable for downstream predictive or psychometric modeling. In behavioral health research, this approach is particularly useful for combining multi-rater ordinal data into interpretable domain scores.

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